Viscous drag and terminal speed

LAMMPS dem09-viscous-linear | Official input tagged unvetted | 29 October 2026

PASS	All four runs compare final falling speed with the official asymptotic terminal-speed target.

Official 0.5% terminal-speed assertion at 0.08 s

Run	Final speed -vz (m/s)	Error (%)	Final z (m)	Official
CPU dt	0.10268243674	0.0463769	0.0008517475	PASS
GPU dt	0.10268243674	0.0463769	0.0008517475	PASS
CPU dt/2	0.10268155218	0.0472380	0.0008543053	PASS
GPU dt/2	0.10268155218	0.0472380	0.0008543053	PASS

Target V_infinity=mg/gamma=0.102730079772 m/s; abs((-vz)/V_infinity-1) <=0.005. The finite-time exact 0.08 s speed is 0.102680656423 m/s; it is a separate supplemental reference. YAML epsilon=1e-10 is same-code regression precision.

Official inputs and native module mapping

Quantity	Unchanged official value / native mapping
Sphere / initial state	R=50 micrometers; rho=2000 kg/m^3; m=1.047197551e-09 kg. Center=(0,0,0.008) m; initial velocity and spin zero.
Gravity / drag	g=9.81 m/s^2 downward; gamma=1e-7 kg/s. Native particle dampingRate=gamma/m=95.49296586 1/s; rotationalDragCoefficient=0. No contacts or walls.
Time / complete outputs	dt=100/50 us; 800/1600 steps to 0.08 s. Every numerical step including t=0: 801/801/1601/1601 frames, total 4804. Four loadable projects and full result ZIP retained.

Force channel and coefficient verification

Native force contains actual external drag and excludes gravity. The module applies F_drag=-m*dampingRate*v_assembly=-gamma*v_assembly. Its input is a rate in 1/s, separate from dimensionful gamma and contact restitution damping. No coefficient was fitted or solver source changed.

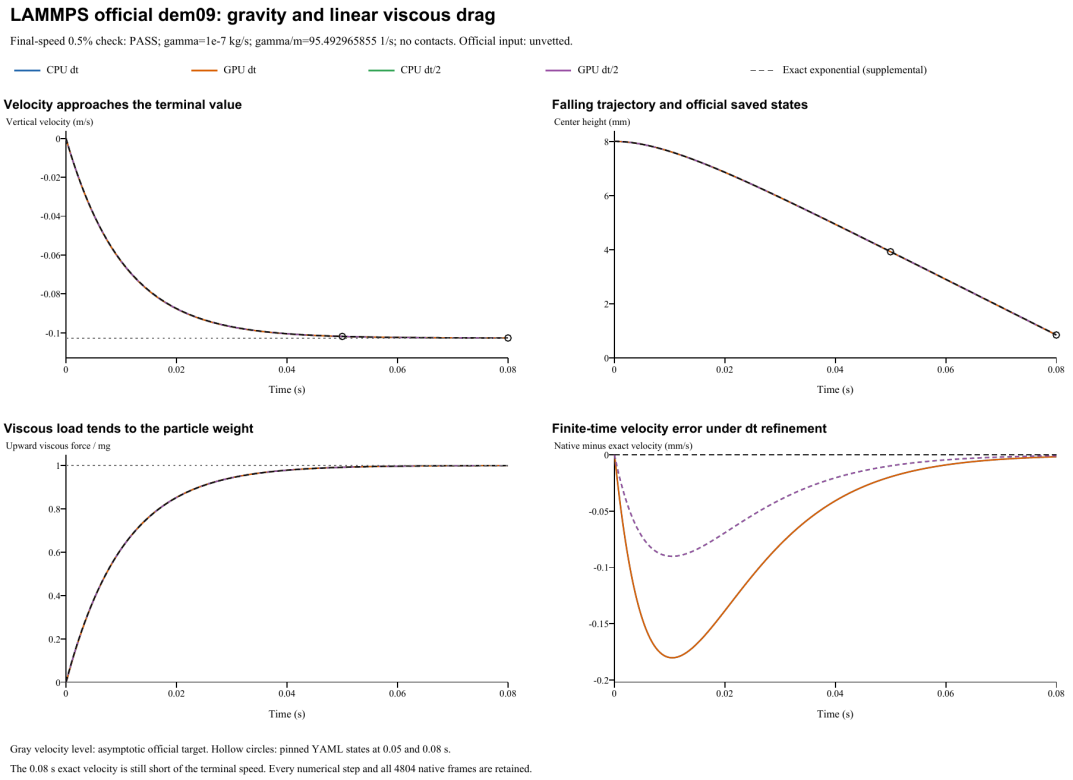
Remove the last half-kick before inferring gamma: v_assembly=v_saved-(dt/2)(F_native/m+g_vector). Check F_native=-gamma*v_assembly; omit near-zero denominators only from coefficient division. Transverse motion, torque, spin and contact count remain zero.

Finite-time exponential reference and source qualification

lambda=gamma/m; tau=m/gamma=0.01047197551 s.
vz=-V_infinity(1-exp(-lambda*t)); z=z0-V_infinity[t-tau(1-exp(-lambda*t))].
The official assertion uses V_infinity, not this finite-time value. The original input has 500/300 steps; one continuous native 800-step event is used. Retained golden-state differences are supplemental.

Pinned official input | Analytic checker | Source/force-channel audit.
Commit 29dd3fd7642a6fe9e8b4dd99120df4068b221250; YAML blob c1948eb56cac61f1c6159fc060d7d191c1ab897c. Official tag: unvetted. LAMMPS was not rerun locally. This input is outside the unchanged historical 24-input catalog.

Native histories and retained evidence



Drag coefficient, trajectory and dissipation

Run	Inferred gamma (kg/s)	Drag residual (N)	Max finite v error (m/s)	Work-energy error (%)
CPU dt	1e-07	1.7e-24	0.0001803	0.032073
GPU dt	1e-07	1.7e-24	0.0001803	0.032073
CPU dt/2	1e-07	1.7e-24	9.019e-05	0.016036
GPU dt/2	1e-07	1.7e-24	9.019e-05	0.016036

CPU/GPU whole-history maximum position / velocity / force differences are 1.73e-18 m / 2.78e-17 m/s / 3.31e-24 N.

Halving dt reduces whole-history finite-time velocity / position errors by factors 1.99920 / 2.00006. The official asymptotic-speed error increases slightly, from 0.0463769% to 0.0472380%. The references differ: the exact 0.08 s speed remains below its infinite-time limit.

Maximum gravity-plus-drag impulse residual is 8.01e-25 kg m/s. Native force is drag only; gravity must be added for total impulse. Mechanical energy falls through viscous dissipation; energy conservation is not the assertion.

Work-energy diagnostic: E=mv^2/2+mgz; compare E+integral(gamma*v_assembly^2 dt) with E(0). Drag power is nonpositive and mechanical energy decreases; the listed residual includes numerical quadrature. The official check remains terminal speed.

Complete results and scope

All 4804 native frames, run/project manifests, four inputs, every-step CSVs and CPU/CUDA logs are retained. Real SimulationSession uses linked FunDEM 0.4.3 native libraries; installed GUI executable byte identity is not established.

Run manifest | Analysis JSON | Checkpoints | Complete native ZIP | Frame/hash audit

Scope: external linear body drag, gamma-to-rate conversion, terminal speed and finite-time gravity/drag dynamics. Contact forces and rotational damping are inactive. Dem09 is inserted between dem07 and dem13; both PFC cases remain after the numbered LAMMPS series.